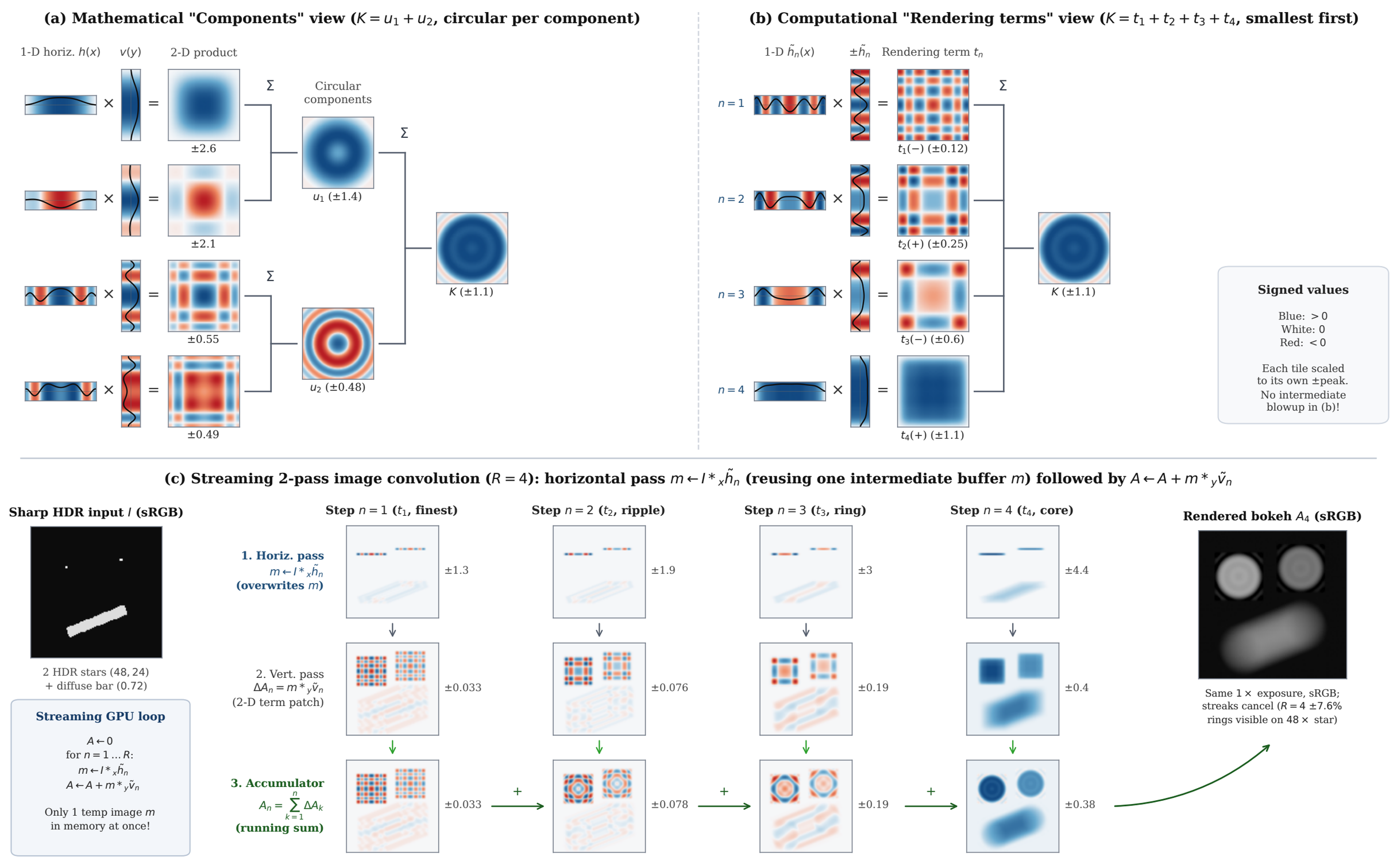

(a) Mathematical "Components" view ($K = u_1 + u_2$, circular per component)
1-D horiz. $h(x)$
$v(y)$
2-D product
Σ
Circular components
±2.6
±2.1
u_1 (±1.4)
±0.55
±0.49
u_2 (±0.48)
K (±1.1)
(b) Computational "Rendering terms" view ($K = t_1 + t_2 + t_3 + t_4$, smallest first)
1-D $\tilde{h}_n(x)$
$\pm\tilde{h}_n$
Rendering term t_n
$n = 1$
$n = 2$
$n = 3$
$n = 4$
$t_1(-)$ (±0.12)
$t_2(+)$ (±0.25)
$t_3(-)$ (±0.6)
$t_4(+)$ (±1.1)
K (±1.1)
Signed values
Blue: > 0
White: 0
Red: < 0
Each tile scaled to its own ±peak.
No intermediate blowup in (b)!
(c) Streaming 2-pass image convolution ($R = 4$): horizontal pass $m \leftarrow I *_x \tilde{h}_n$ (reusing one intermediate buffer m) followed by $A \leftarrow A + m *_y \tilde{v}_n$
Sharp HDR input I (sRGB)
2 HDR stars (48, 24)
+ diffuse bar (0.72)
Streaming GPU loop
$A \leftarrow 0$
for $n = 1 \dots R$:
$m \leftarrow I *_x \tilde{h}_n$
$A \leftarrow A + m *_y \tilde{v}_n$
Only 1 temp image m in memory at once!
Step $n = 1$ (t_1, finest)
Step $n = 2$ (t_2, ripple)
Step $n = 3$ (t_3, ring)
Step $n = 4$ (t_4, core)
1. Horiz. pass
$m \leftarrow I *_x \tilde{h}_n$
(overwrites m)
2. Vert. pass
$\Delta A_n = m *_y \tilde{v}_n$
(2-D term patch)
3. Accumulator
$A_n = \sum_{k=1}^{n} \Delta A_k$
(running sum)
±1.3
±1.9
±3
±4.4
±0.033
±0.076
±0.19
±0.4
±0.033
±0.078
±0.19
±0.38
+
Rendered bokeh A_4 (sRGB)
Same 1× exposure, sRGB;
streaks cancel ($R = 4$ ±7.6%
rings visible on 48× star)